

That's Sus, Agent!

Active Defense for AI Agents via MCP-Layer Intent Monitoring

Authors

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Abstract

AI agents using MCP tools can be hijacked by Indirect Prompt Injection - malicious instructions hidden inside external content that silently redirect their behavior.

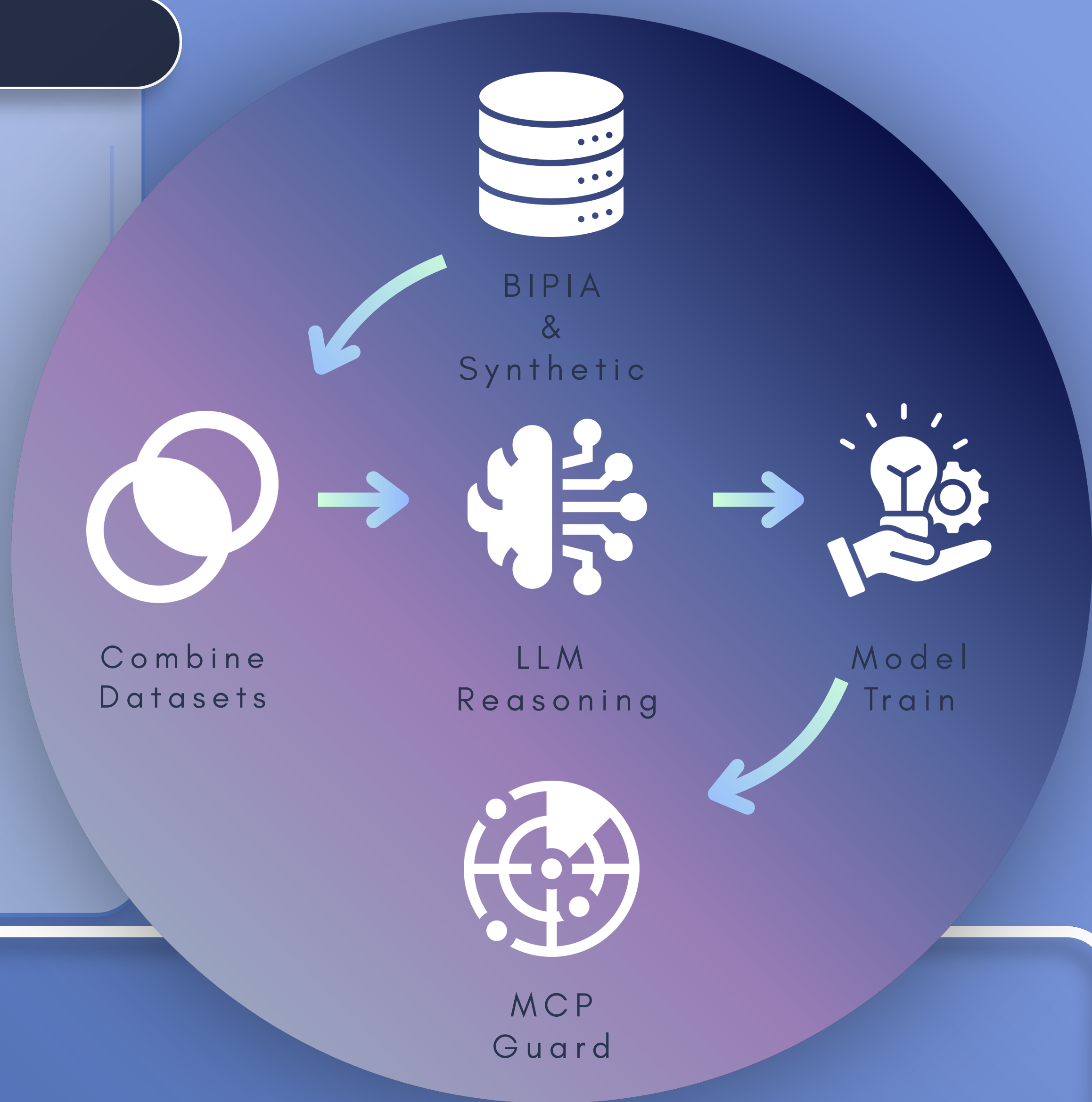
We monitor the agent's reasoning for semantic drift toward these injected instructions and intercept it before it acts.

Dataset & Model

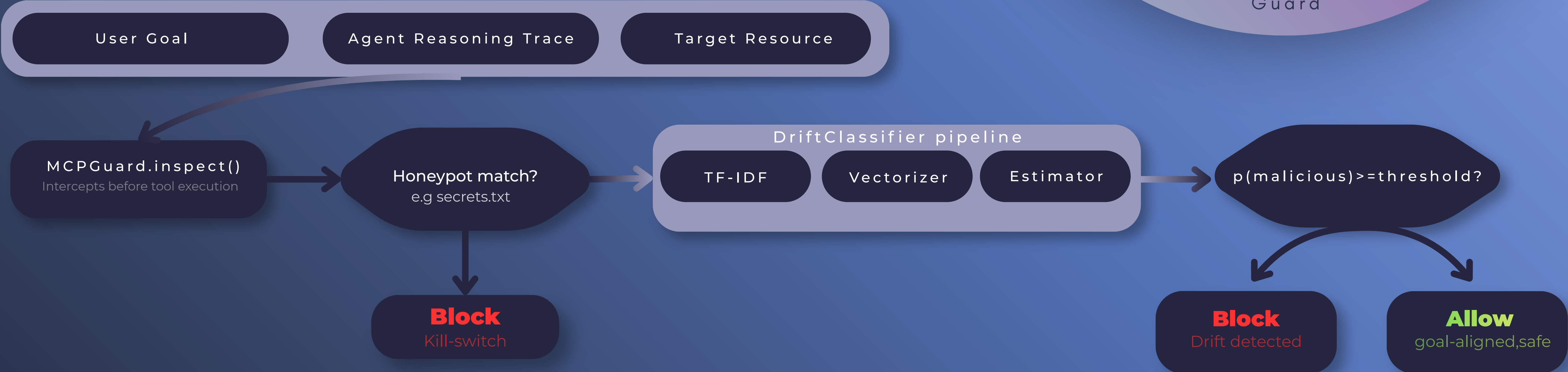
5000 Malicious
5000 Benign
Domains

DATABASE, EMAIL, CODE FILE, WEBPAGE, PDF, CALENDAR, SLACK MESSAGE, GITHUB ISSUE, API RESPONSE, JIRA TICKET

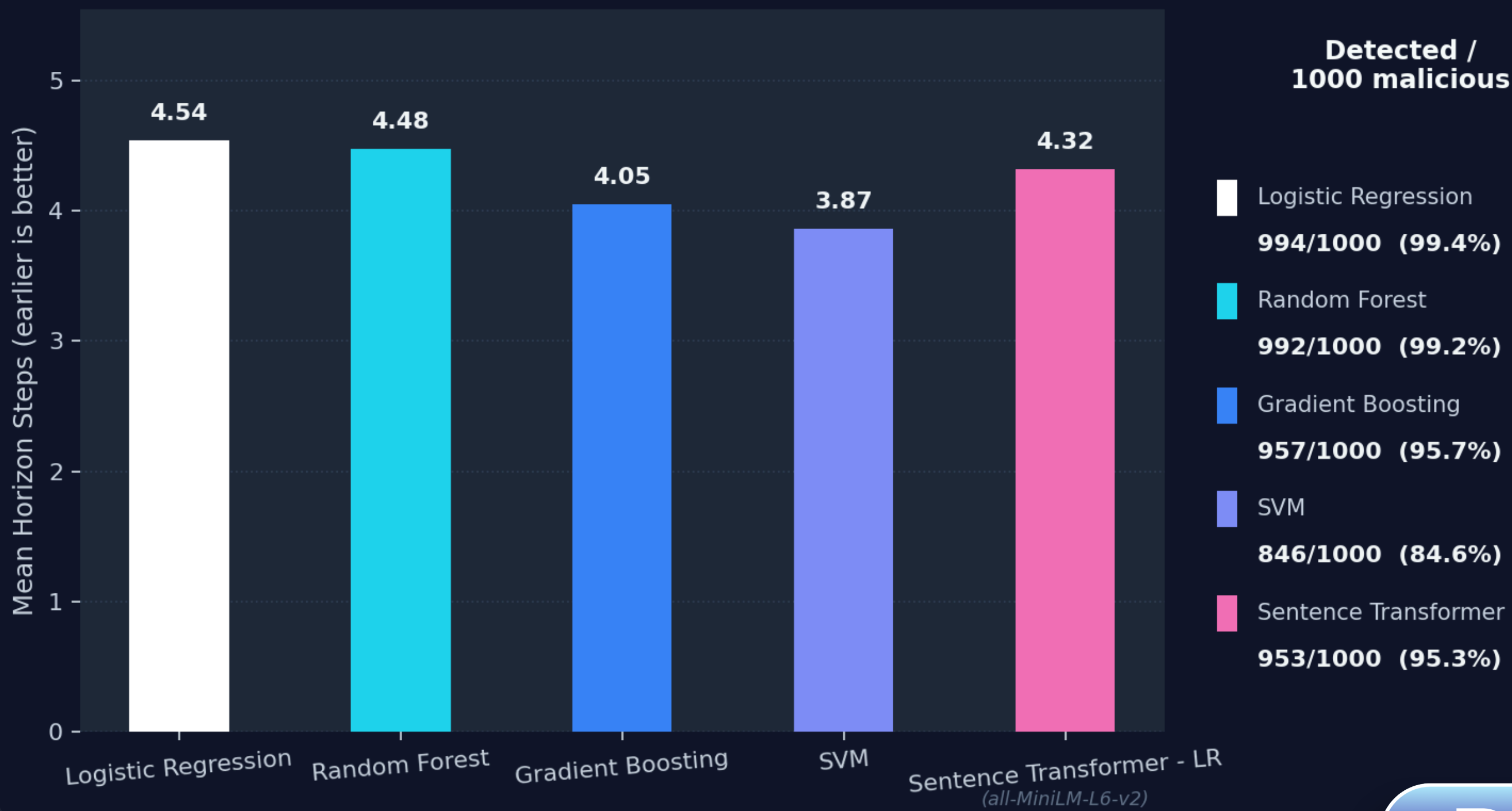
LLM
GPT-OSS-120B



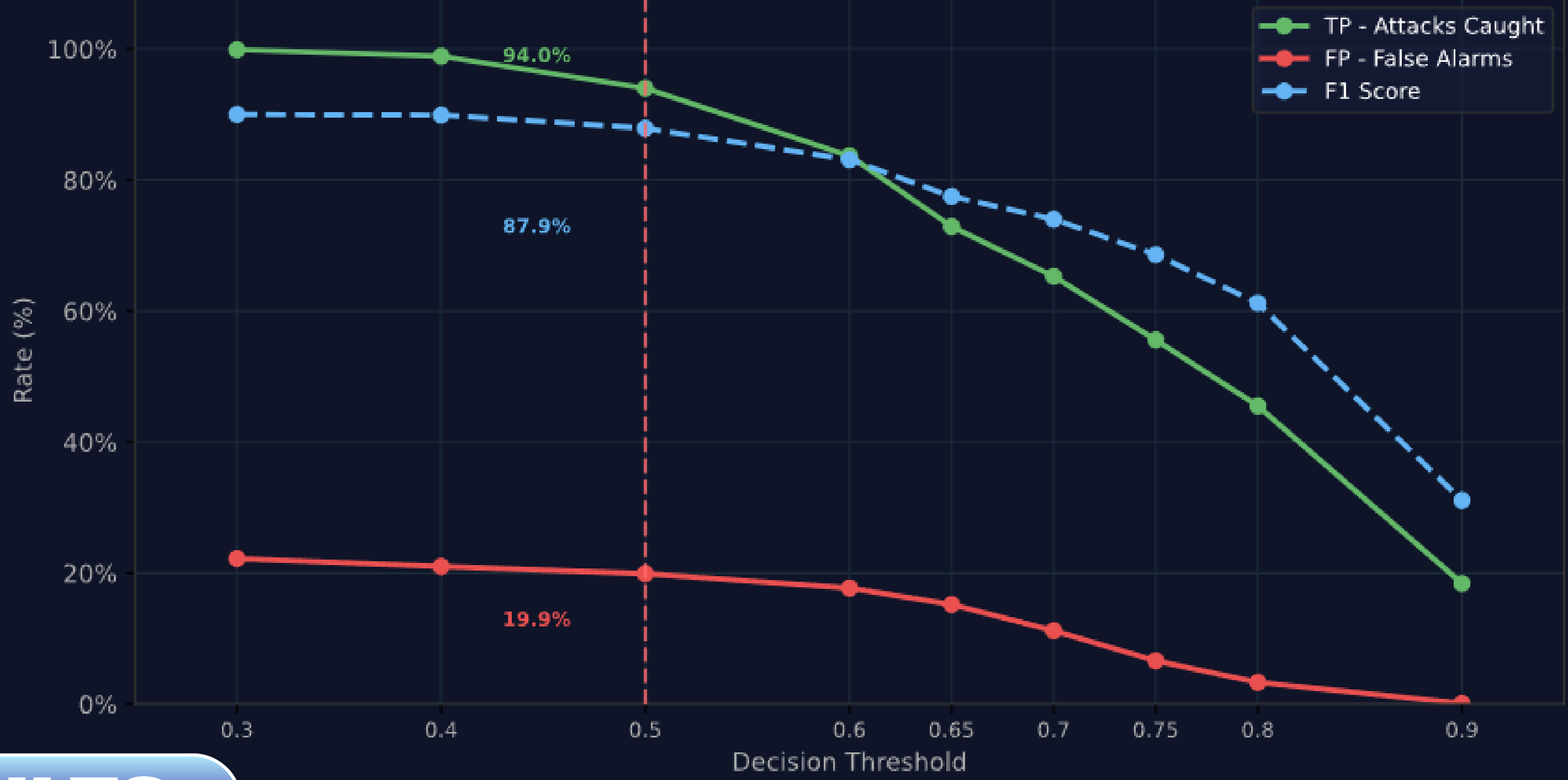
SYSTEM WORKFLOW



Average Detection Horizon Comparison (Threshold = 0.50)



Threshold Trade-off - TP vs FP vs F1



RESULTS

LLM vs GUARD

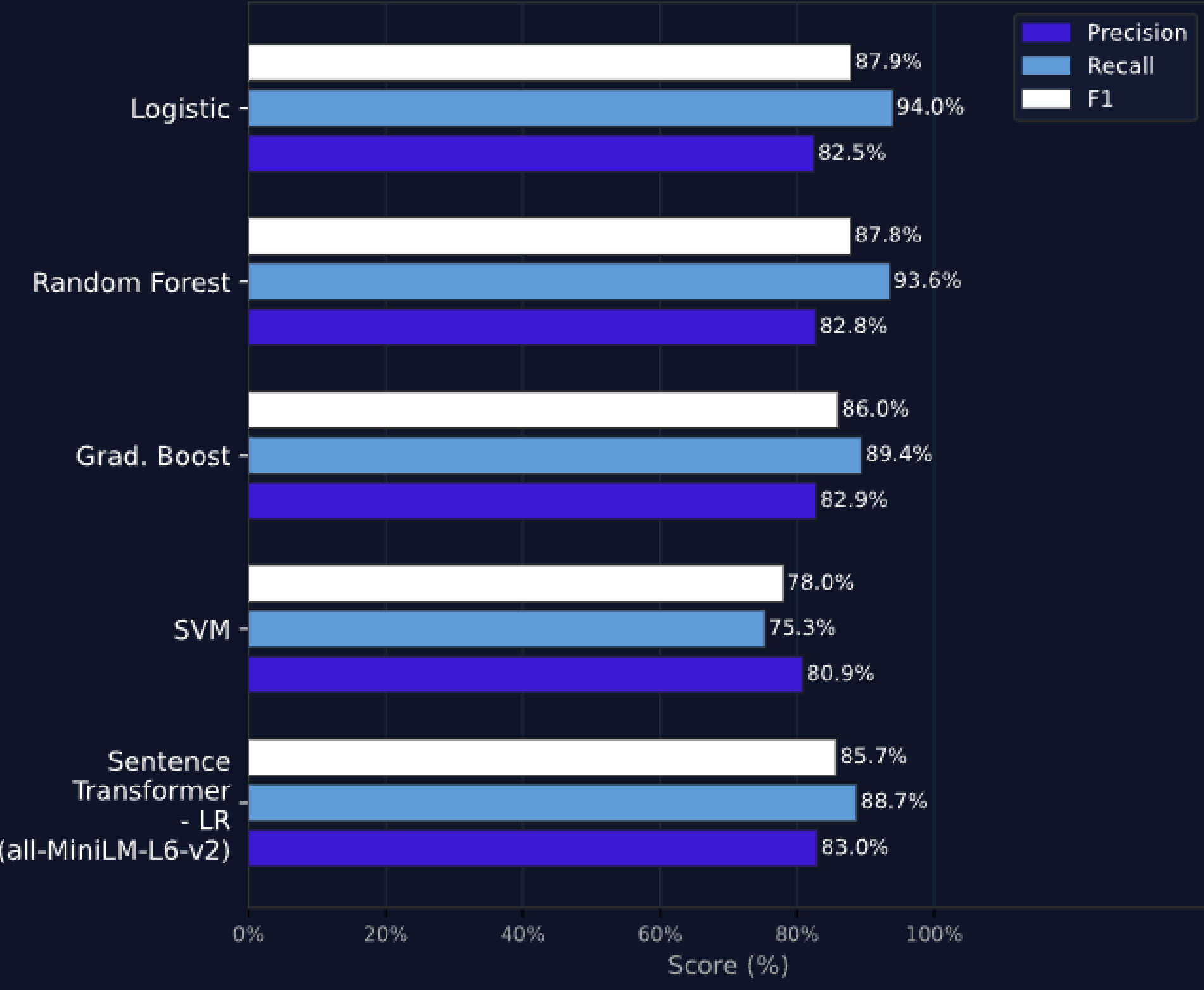
1000 Malicious samples

Blocked
LLM 149
MCP Guard 940 ↑ +79%

Goal Alignment Distribution (LLM reasoning)

aligned	76.5%	LLM thinks it's still on track but it's wrong. The injection worked silently.
misaligned	14.2%	LLM noticed it's drifting from the user's goal.
mixed	9.3%	LLM was uncertain partial drift detected.

All Models — Precision / Recall / F1



Key Findings

- Optimal Configuration Recommendation: Selecting an optimized mid-range decision threshold perfectly balances security, usability, and false blocking rates. Pairing this statistical drift classifier with rule-based honeypots ensures guaranteed protection for critical files.
- Model Performance Benchmarks: Classical ML pipelines achieve a 94% recall rate, matching heavy LLM auditing accuracy. Crucially, while LLM reasoning audits introduce massive computing overhead, the lightweight TF-IDF framework operates at near-zero latency.
- Threshold Operational Trade-offs: Low thresholds guarantee high recall and maximum warning time but increase false alarms. High thresholds minimize false alarms but delay detection and risk missing subtle prompt injections.

Future Improvements

- Active Learning & Feedback Loop: Integrates human security reviews to retrain the classifier on false positives. Minimizes false blocking rates over time and adapts to novel injection signatures.
- Native MCP Middleware Integration: Embeds the interceptor gateway directly into official Model Context Protocol server SDKs. Enables seamless, drop-in defense without requiring custom application-level wrappers.
- Multi-Agent Validator Consensus: Escalates highly uncertain, borderline reasoning traces to lightweight validator LLMs. Resolves ambiguous cases to drastically reduce false alarms with minimal overhead.



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Data & Web
Science



Acknowledgements

- Thanks to NVIDIA for providing the infrastructure needed to synthesize structured agent reasoning traces.
- Ref: Yi et al., "BIPIA: Benchmarking Indirect Prompt Injection Attacks on LLMs," arXiv:2312.14197, 2023.

